RSU-101U Specification

V2X Roadside Communication Unit , IEEE 1609.x protocol stack



Overview:

RSU-101U is a V2X roadside communication unit designed to enable DSRC V2X on the existing traffic controller system. Preloaded IEEE 1609.2/3/4 stack running on ThreadX in an IP67 enclosure, RSU-101U can co-operatively work with the traffic controller unit to receive and forward SPaT/MAP messages as the payload for V2I/I2V communication between vehicles and traffic controllers in ITS applications.

Optimized DSRC V2X system supports innovative DSRC Vehicle to Infrastructure (V2I) and Vehicle to Vehicle (V2V) applications to improve road safety and reduce congestion in smart city.

Key Features:

System:

- Complete DSRC V2X elements of communication processor, DSRC RF, GNSS positioning, and hardware security in an IP67 enclosure enables C-ITS applications.
- Preloaded firmware of IEEE 1609.2, 1609.3, 1609.4 protocol stack and SAE J2735 message set on RSU-101U. RSU-101U receives and forwards the SPaT/MAP messages as the payload for V2I/I2V communication between vehicles and traffic controllers in ITS applications.
- Autotalks® Craton 3-core communication processor, Pluton RF transceiver, Telit® SL869 GNSS receiver, Infineon® SLE97 HSM, 128MB DDR3, and 32MB NOR in a standalone box, no external CPU and GNSS required.
- High-performance Telit® SL869 GNSS receiver supports GPS/Glonass/Galileo/QZSS constellations and provides accurate positioning.
- Infineon® SLE97 Hardware Security Module (HSM) provides ECDSA 256 signing and secure private key storage in a tamper proof device.
- 802.3at PoE (48V/25W) or DC 6-32V power input provides application flexibility.
- Extended temperature -40°C ~ +70°C design and components (Infineon SLE97® -20°C ~ +85°C now) ensure environmental reliability.
- Two outdoor 7.6dBi DSRC antennas enable robust performance in harsh weather condition.
- One outdoor GNSS antenna.

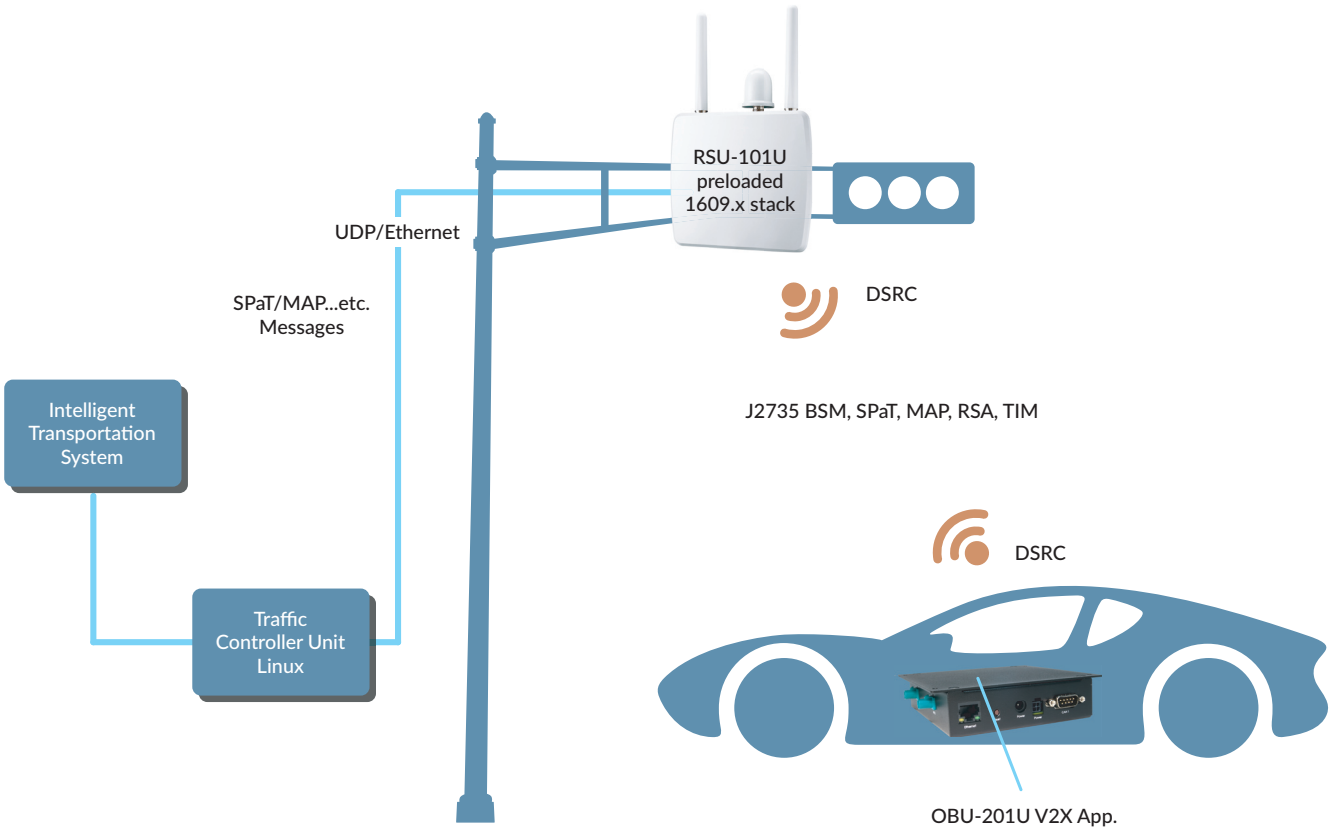
Performance:

- More than +20dBm Class C RF spectrum mask compliant with margins.
- Superior fading sensitivity in the typical ETSI defined C2C multipath scenarios increases wireless coverage.
- Internal 40MHz BW filter provides immunity to out-of-band radio interferences in multi-channel operation.
- Dual DSRC PHY supports concurrent dual channel operation.
- Dedicated oscillator provides high frequency accuracy at ±6.0ppm in –40°C ~ +70°C temperature range.
- Dynamic and accurate power control in a wide 4.5dBm ~ 25dBm output range provides superior performance stability.
- GNSS positioning supports 1.5m accuracy (CEP50 with SBAS) and 10Hz refresh rate.
- Hardware verification engine in Craton supports over 2,000 ECDSA 256-bit verifications per second.
- HSM supports less than 50ms latency on ECDSA 256-bit signing.
- High ESD protection design ensures immunity and robustness of all ports in ESD events. (IEC 61000-4-2 Level 4, Contact ±8kV, Air ±15kV)

Flexibility:

- External GNSS support through RS-232 provides high application flexibility, in addition to default on-board Telit GNSS module.
- 802.3at PoE or DC 6 -32V dual power supply provides high application flexibility.
- Module-structured architecture provides high application flexibility to easily customize PCB-201 carrier board on the same architecture and software environment, no impact on V2X protocol stack and DSRC performance in VTX-201 sub-system.

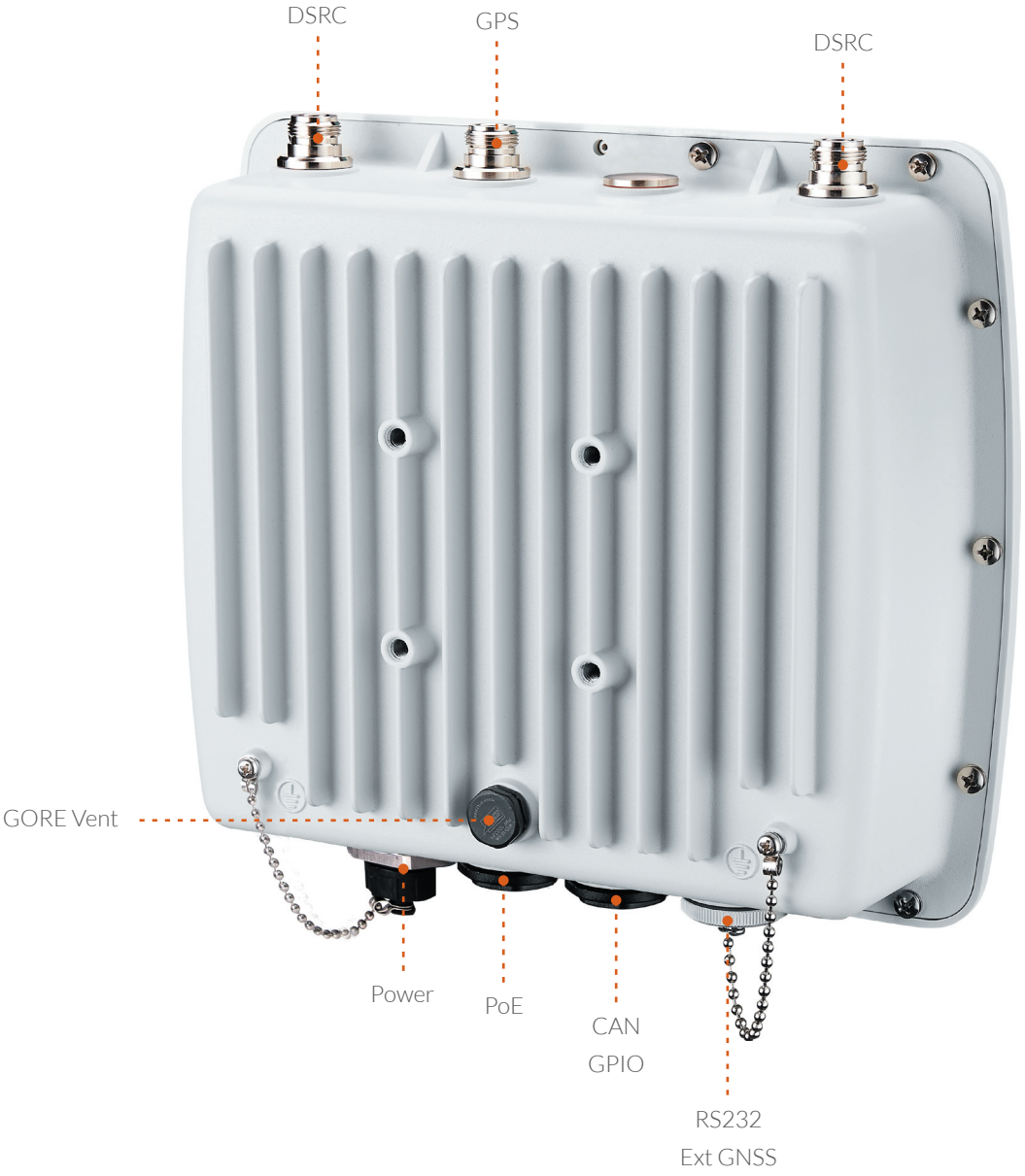
Typical Using Case:



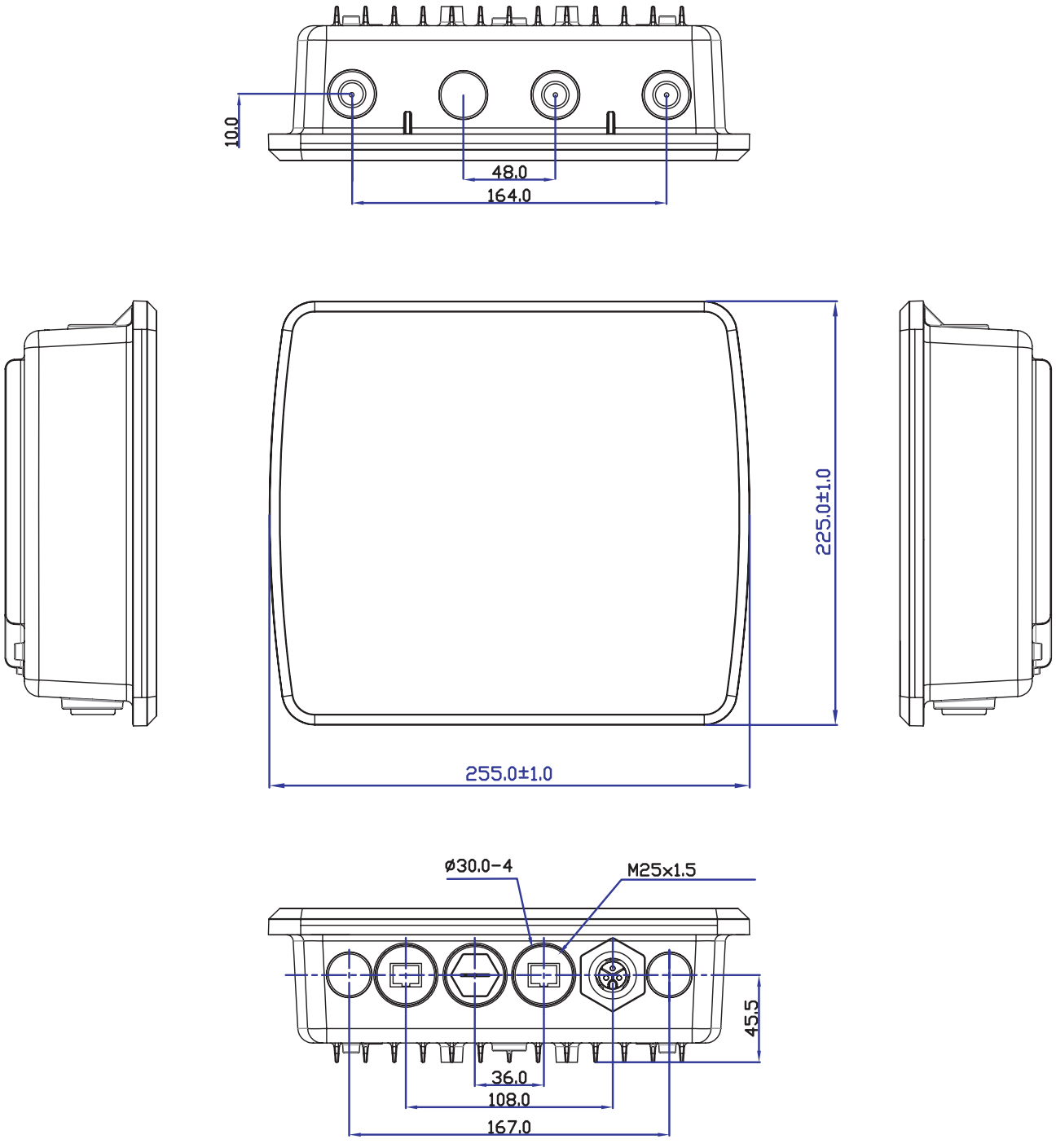
Remark:

The traffic controller unit acts as SPaT/MAP generator to perform the relative encode/decode. RSU-101U serves to receive and forward the SPaT/MAP messages as a payload for communication between vehicles and traffic controllers in co-operative ITS. V2I/I2V applications can be flexibly implemented on the traffic controller unit or RSU-101U.

Connectivity:

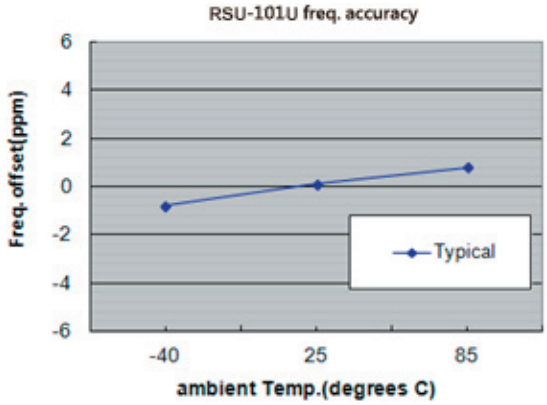


Mechanical Outline:

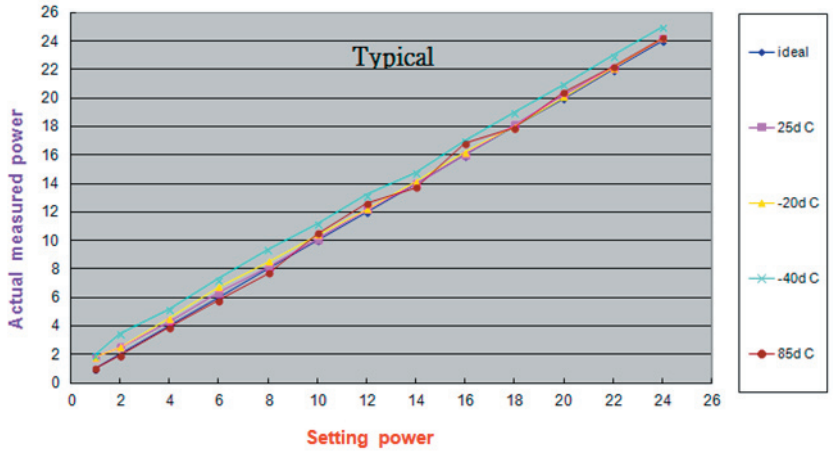


Critical Facts of RSU-101U

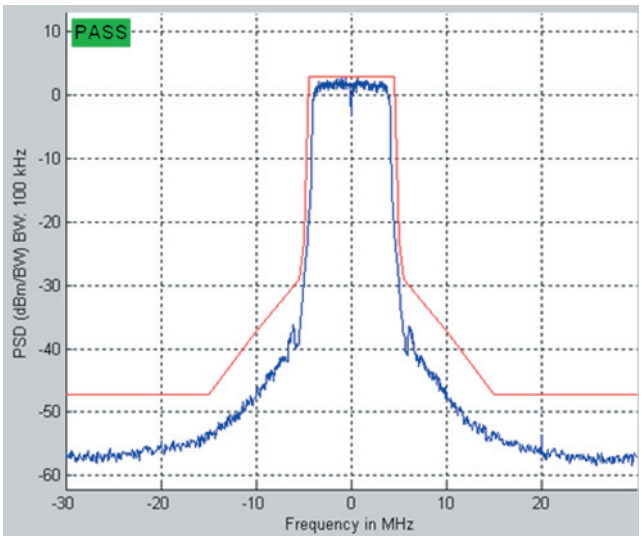
1. Frequency Accuracy:



2. Power Control Accuracy:



3. Class C mask performance at +20dBm from -40°C ~ +85°C:



4. Consistent EVM performance over a wide 5~22dBm power levels and -40°C~+85°C temperature range provide high channel efficiency.

	25 degrees C		85 degrees C		-40 degrees C	
	TX2	TX1	TX2	TX1	TX2	TX1
dut1	-28	-28	-29	-27.8	-27.7	-27.5
dut2	-30.5	-27.5	-30	-27	-30	-27.4
dut3	-31.4	-26.3	-30.2	-28.8	-28	-27

Specifications:

Software:

Operation System	ThreadX RTOS
Firmware	Preloaded standard compliance stack and firmware including: » IEEE 802.11p - 2010 » IEEE 1609.2 - 2016 » IEEE 1609.3 - 2016 » IEEE 1609.4 - 2016 » SAE J2735_201603 » uCoupler: WSM-UDP converter
Development Tool	1. Unex's SDK (contains header files (.h), archived object codes (.a), example codes (.c), makefile, API guide, environment setup guide, quick start guide) 2. GCC cross compiler to compile ThreadX OS image under Linux 3. Hardware Guide 4. User's Guide
System Service	RS-232 console port (baud rate 115200 bps)

Hardware:

Chipset

- Autotalks® CRATON V2X communication processor, three 240MHz CPU cores
- Autotalks® PLUTON V2X RF Transceiver
- Telit® SL869 GNSS module
- Infineon SLE97 Hardware Security Module (HSM)

System Memory

32MB NOR, 128MB DDR3

DSRC

- frequency band: 5.85 ~ 5.925 GHz (ITS-DSRC)
- radio mode: 802.11p, ITS-G5
- channel: 172, 174, 176, 178, 180, 182, 184
- channel bandwidth: 10MHz (5MHz & 20MHz by project)
- data rate: 3, 4.5, 6, 9, 12, 18, 24, 27Mbps for 10MHz BW signal
- RF transmit power: > +20dBm, Class C RF spectrum mask compliant with margins.
- frequency accuracy: ± 6.0ppm

DSRC Static Sensitivity (Typical)

Conditions	-40°C	+25°C	+85°C
-92dBm	3Mbps	-97Mbps	-93Mbps
-92dBm	4.5Mbps	-97Mbps	-93Mbps
-91dBm	6Mbps	-95Mbps	-91Mbps
-89dBm	9Mbps	-93Mbps	-89Mbps
-85dBm	12Mbps	-90Mbps	-86Mbps
-83dBm	18Mbps	-86Mbps	-83Mbps
-75dBm	24Mbps	-80Mbps	-75Mbps
-73dBm	27Mbps	-78Mbps	-74Mbps

DSRC Fading Sensitivity	Power @ 10% PER sensitivity (6Mbps, 1000B packet), fading channels as 5 typical C2C multipath scenarios defined by ETSI: ± 2dBm » Rural LOS: -92.5dBm » Highway LOS: -91.5dBm » Urban Approaching LOS: -91.5dBm » Crossing NLOS: -89.5dBm » Highway NLOS: -88.5dBm
GNSS	» refresh Rate: 10Hz » sensitivity: -135dBm » NMEA Standard: NMEA 0183
GNSS Accuracy	1.5m (CEP50 with SBAS)
Hardware Security	» ECDSA 256-bit signing (<50ms) » private key and public key verification (over 2,000 ECDSA 256-bit verifications per second) » secure storage of private keys in HSM
External Connector	» two DSRC antenna connectors » one GPS antenna connector » one standard DC 6-32V power » one 48V PoE » one CAN and GPIO connector » one RS232 (RJ-45, EIA-561) for external GNSS
Antenna	» two outdoor 7.6dBi DSRC antennas » one outdoor GNSS antenna

Power Supply

dual power input:

1. 802.3at PoE (48V/25W)
2. DC 6-32V

Operation Voltage

48V on PoE or DC 6-32 V

Power Consumption

12V power input			
Condition	**25°C**	**85°C**	**-40°C**
Tx @ 20~8dBm (RF duty cycle = 7%)	0.3A	0.315A	0.3A
Idle	0.29A	0.3A	0.29A

Operation Temperature Range

ambient: -40°C~ +70°C

(Infineon SLE97 Hardware Security Module supports -20°C ~ +85°C)

Operating Humidity

10% ~ 95%, non-condensing

Storage Humidity

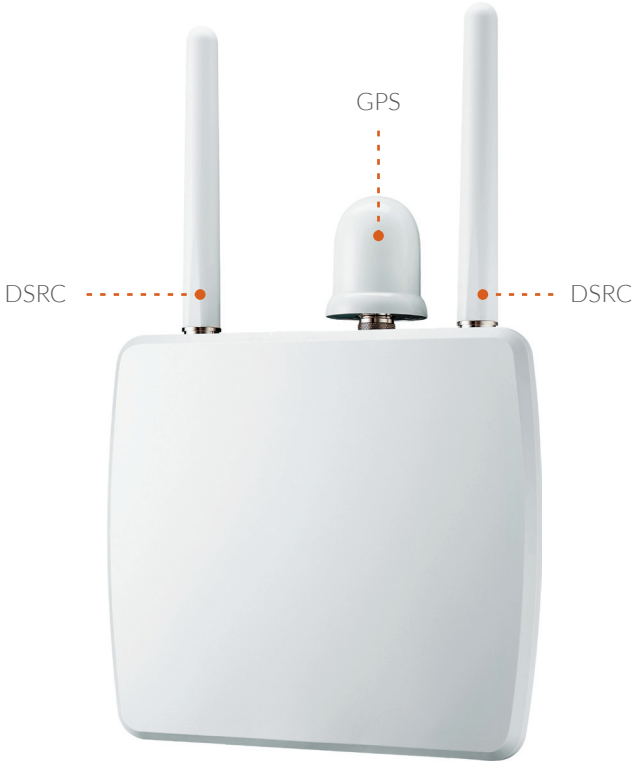
max. 95%, non-condensing

Product Dimension

225mm (L) x 255mm (W) x 78mm (H)

Package Contents:

1. RSU-101U x 1 with preloaded IEEE 1609.x stack running on ThreadX
2. Two DSRC outdoor antennas
3. One outdoor GNSS antenna
4. Mounting Bracket with two clamp and needed accessories
5. Unex's SDK, Hardware Guide and User's Guide downloaded from Unex's server.





Ordering Information:

RSU-101U	V2X Roadside Communication Unit, 1609.x protocol stack
RSU-101E	V2X Roadside Communication Unit, ETSI TC-ITS protocol stack
RSU-101	V2X Roadside Communication Unit

Recommended Software Option:

MISRA C: 2012 compliance facilitates software code to ensure safety, portability and reliability in Unex's V2X embedded systems.

VAS-201x	CAMP VSC-A or ETSI TC-ITS Day-One or field V2X applications
VVT-201	V2X Diagnostic Firmware
V2X Turn-Key Pilot Deployment	by project request

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